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(54) **COMPACT PHASE SHIFTER LAYOUT**

(71) Applicant: **Apple Inc.**, Cupertino, CA (US)

(72) Inventors: **Bo Yu**, San Diego, CA (US); **Seyedmohammadreza Razavian**, San Diego, CA (US); **Zhengan Yang**, Del Mar, CA (US); **Haitao Cheng**, San Jose, CA (US); **Ali Parsa**, Poway, CA (US); **Zhang Jin**, San Diego, CA (US)

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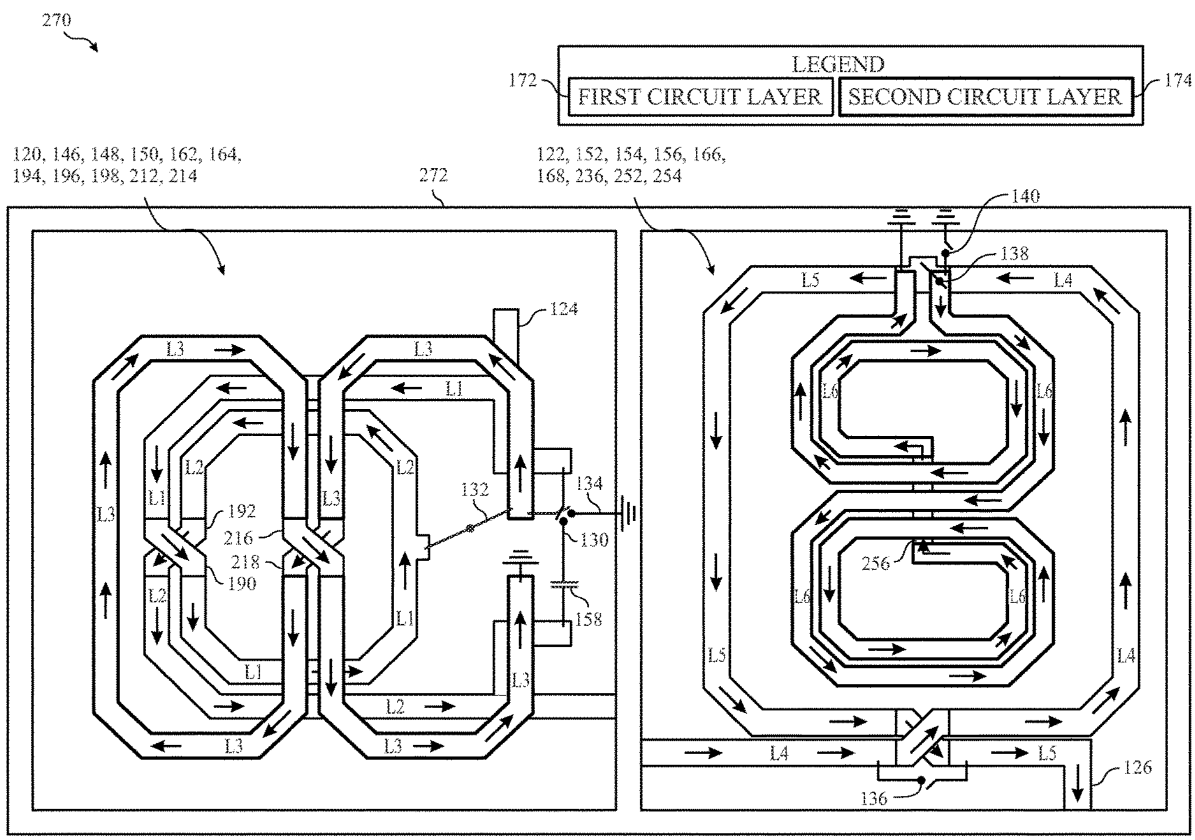
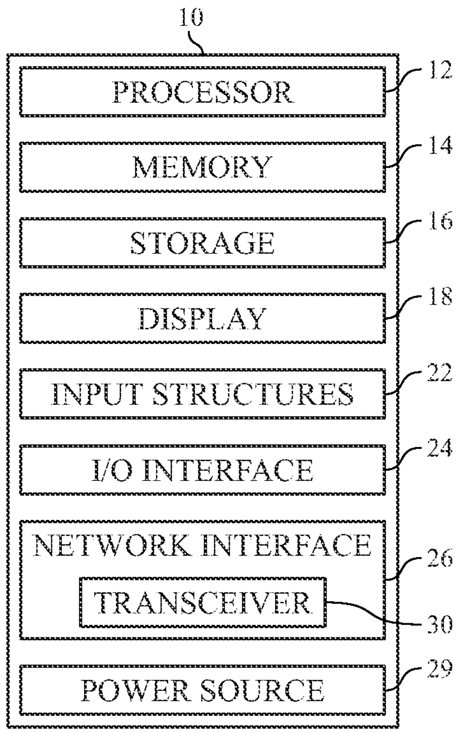
(57) **ABSTRACT**

This disclosure is directed to phase shifter circuitry with reduced insertion loss and/or reduced area compared to other phase shifter circuitry. The phase shifter circuitry may include a first phase shifter circuit and a second phase shifter circuit. The first phase shifter circuit may include two overlaid coils forming three inductors. Similarly, the second phase shifter circuit may include two overlaid coils forming three inductors. A coil of the first phase shifter circuit may be extended orthogonally to a coil of the second phase shifter. As such, the first phase shifter circuit and the second phase shifter circuit may have reduced undesired induced currents during operation. Moreover, the coils of each of the phase shifter circuits are disposed adjacently on multiple circuit layers to improve the insertion loss of the phase shifter circuitry, reduced the area occupied by the phase shifter circuitry, or both, among other things.

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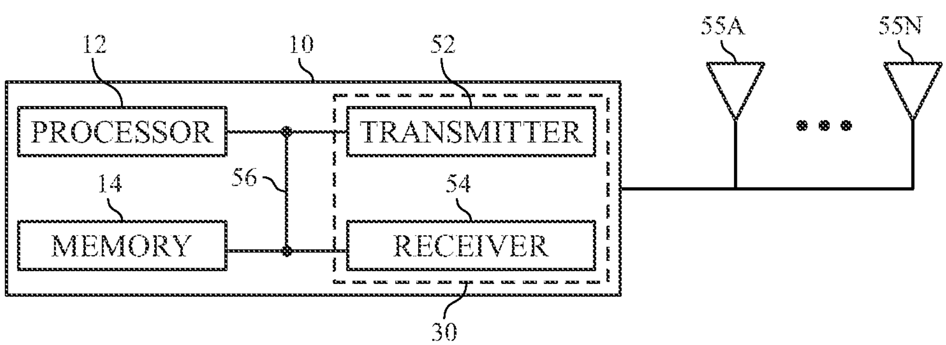
PROCESSOR
12
MEMORY
14
STORAGE
16
DISPLAY
18
INPUT STRUCTURES
22
I/O INTERFACE
24
NETWORK INTERFACE
26
TRANSCEIVER
30
POWER SOURCE
29

FIG. 1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10
52
55A
55N
PROCESSOR
TRANSMITTER
56
14
54
MEMORY
RECEIVER
30

FIG. 2

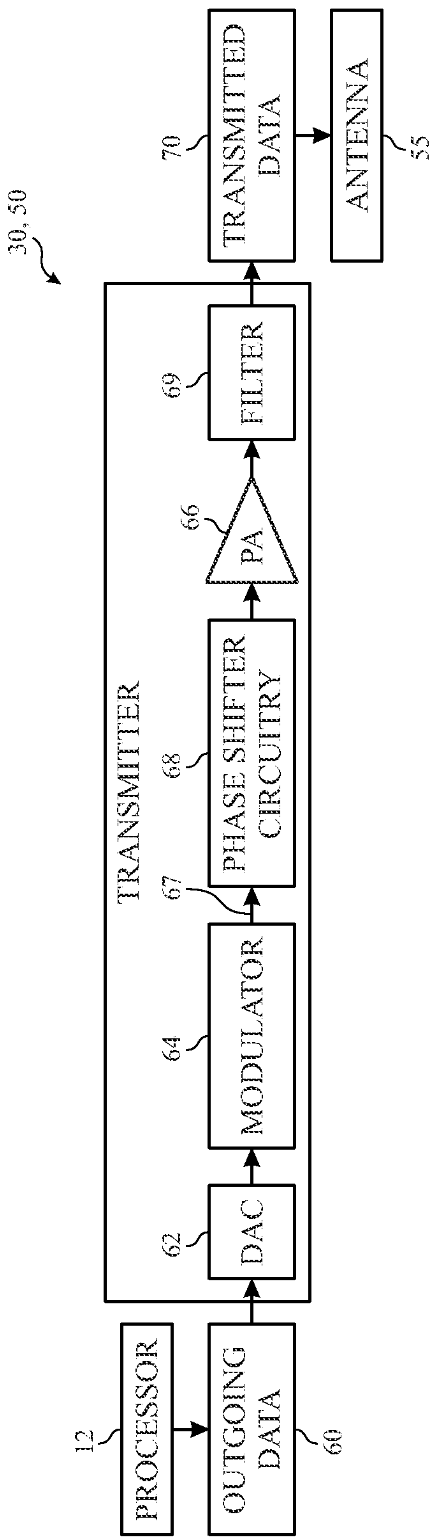

30, 50
PROCESSOR
12
OUTGOING DATA
60
TRANSMITTER
DAC
62
MODULATOR
64
67
PHASE SHIFTER CIRCUITRY
68
PA
66
FILTER
69
TRANSMITTED DATA
70
ANTENNA
55

FIG. 3

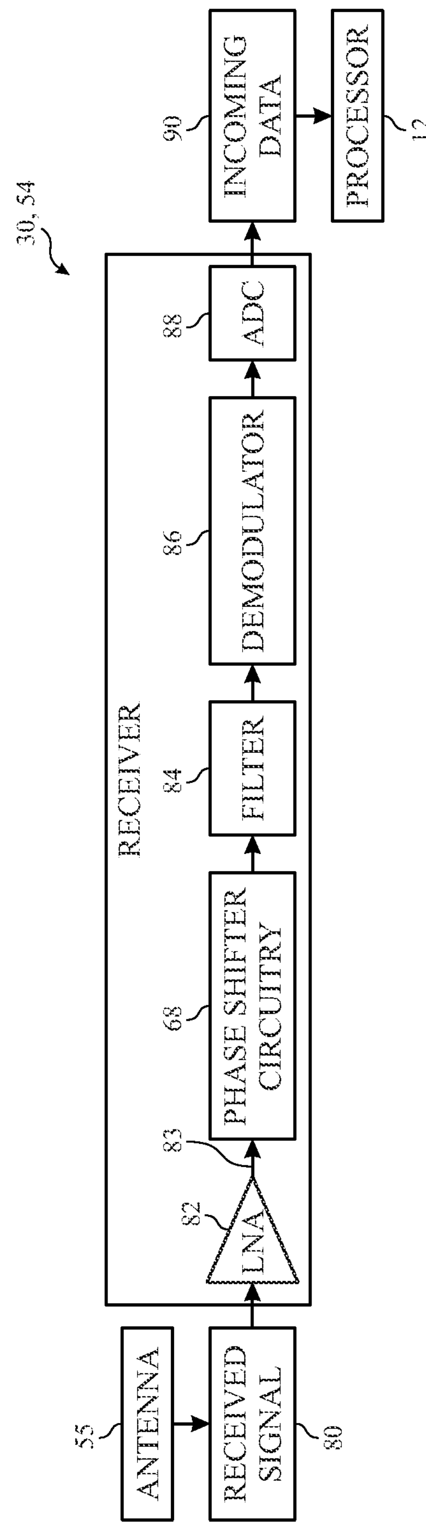

30, 54
ANTENNA
55
RECEIVED SIGNAL
80
RECEIVER
LNA
82
83
PHASE SHIFTER CIRCUITRY
68
FILTER
84
DEMODULATOR
86
ADC
88
INCOMING DATA
90
PROCESSOR
12

FIG. 4

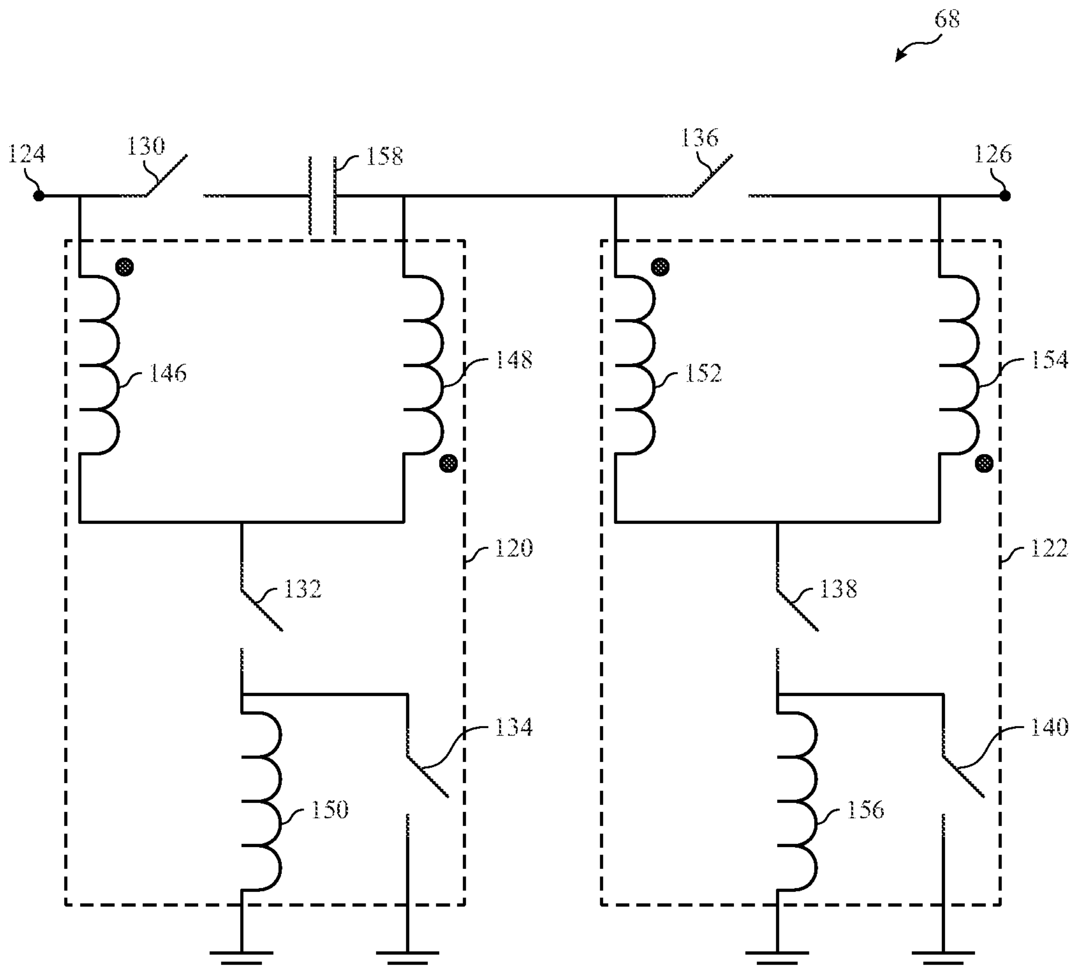

68
124
130
158
136
126
146
148
152
154
120
122
132
138
134
140
150
156

FIG. 5

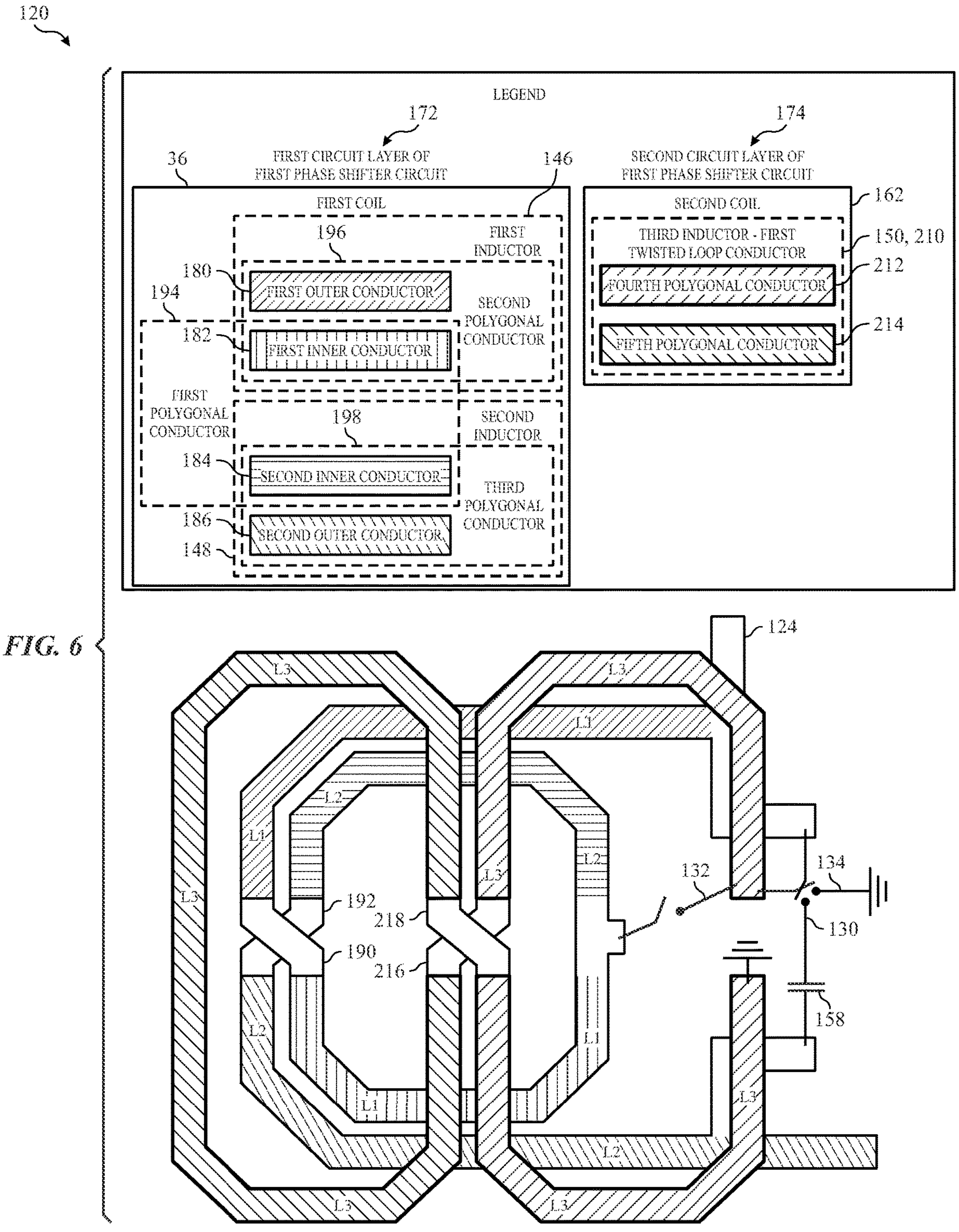

120
LEGEND
172
FIRST CIRCUIT LAYER OF FIRST PHASE SHIFTER CIRCUIT
174
SECOND CIRCUIT LAYER OF FIRST PHASE SHIFTER CIRCUIT
36
146
FIRST COIL
FIRST INDUCTOR
196
180
FIRST OUTER CONDUCTOR
SECOND POLYGONAL CONDUCTOR
194
182
FIRST INNER CONDUCTOR
FIRST POLYGONAL CONDUCTOR
198
SECOND INDUCTOR
184
SECOND INNER CONDUCTOR
THIRD POLYGONAL CONDUCTOR
186
SECOND OUTER CONDUCTOR
148
SECOND COIL
162
THIRD INDUCTOR - FIRST TWISTED LOOP CONDUCTOR
150, 210
FOURTH POLYGONAL CONDUCTOR
212
FIFTH POLYGONAL CONDUCTOR
214
124
L3
L1
L2
132
134
192
218
190
216
130
158

FIG. 6

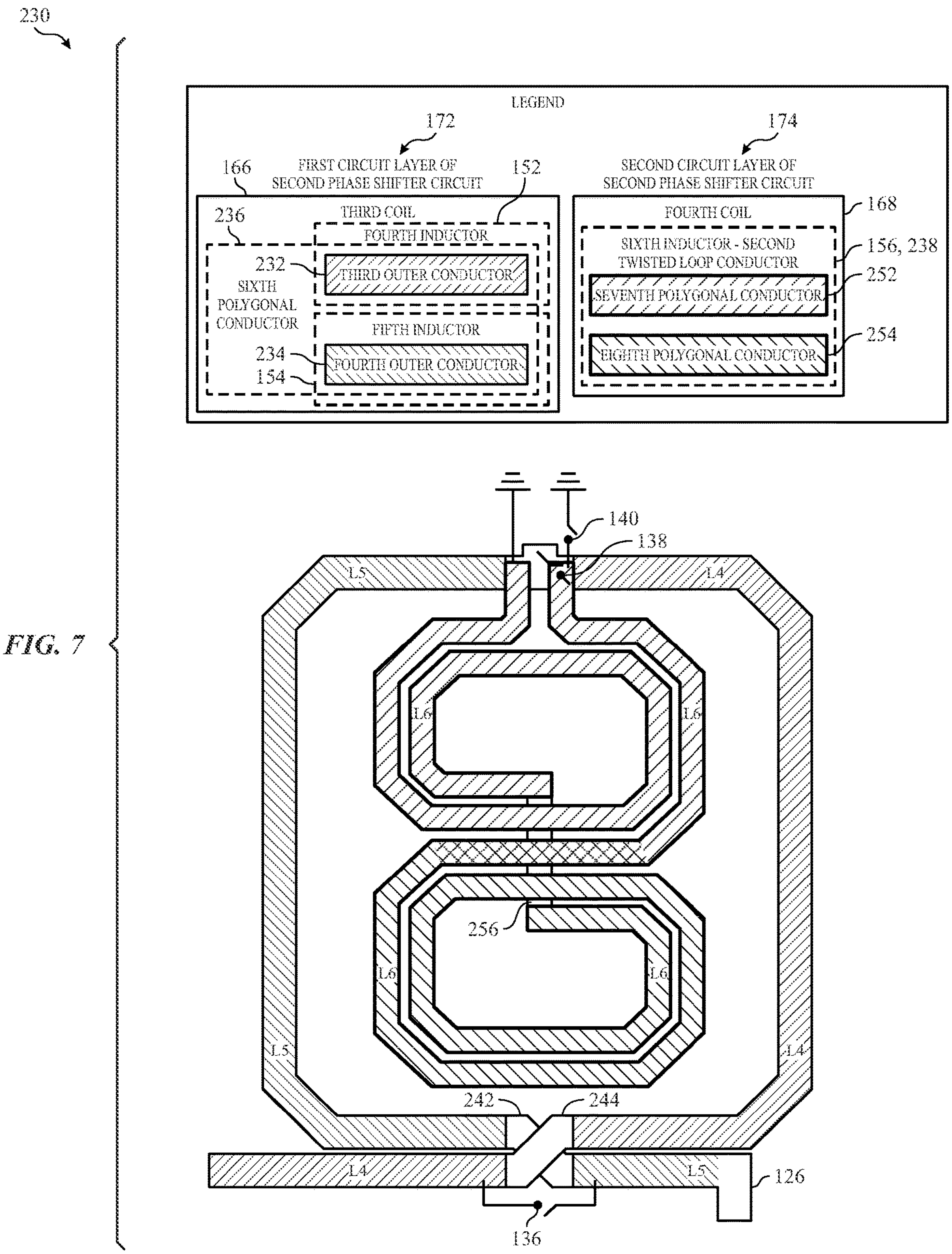
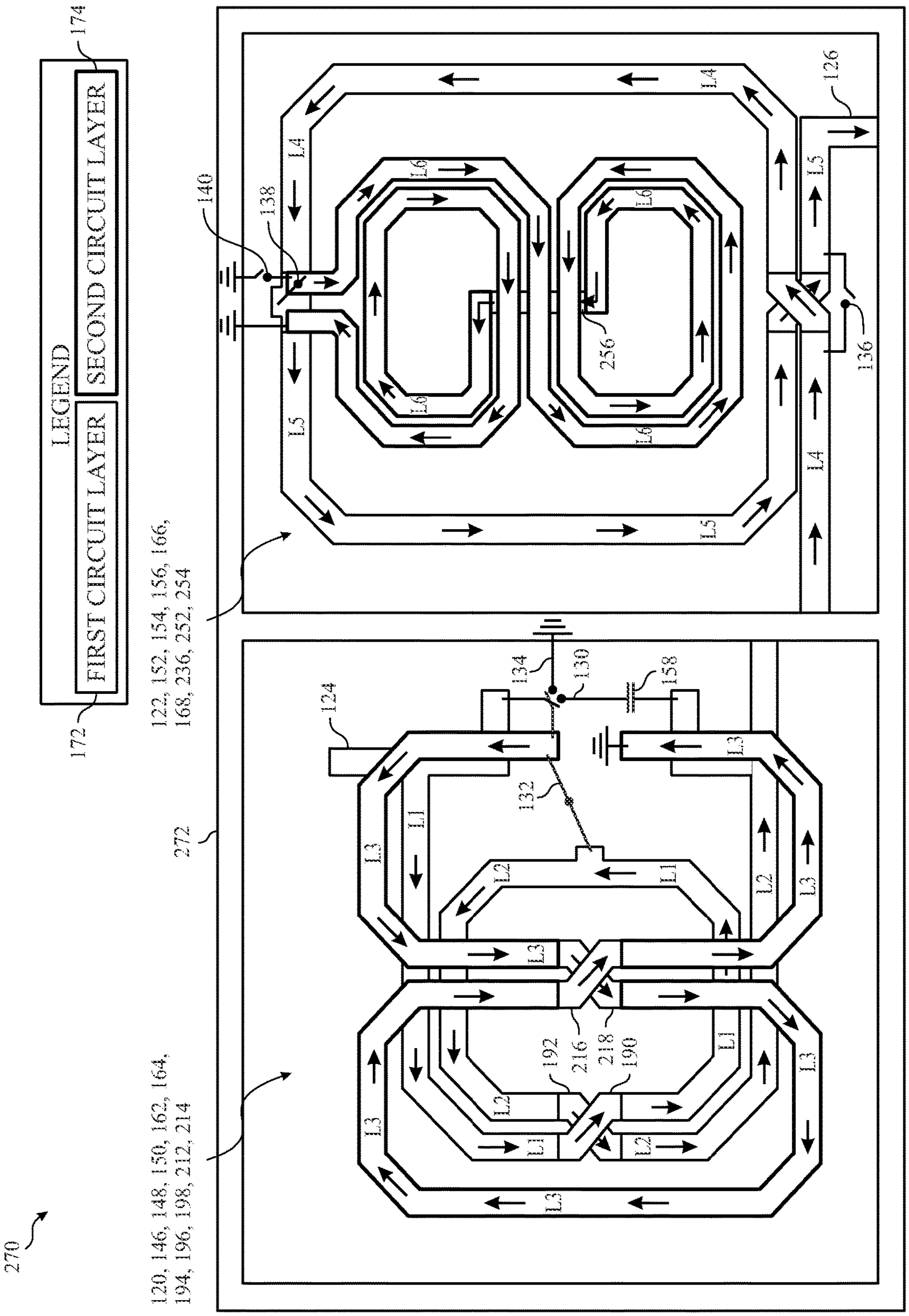

230
LEGEND
172
FIRST CIRCUIT LAYER OF SECOND PHASE SHIFTER CIRCUIT
174
SECOND CIRCUIT LAYER OF SECOND PHASE SHIFTER CIRCUIT
166
152
236
THIRD COIL
FOURTH INDUCTOR
232
THIRD OUTER CONDUCTOR
SIXTH POLYGONAL CONDUCTOR
FIFTH INDUCTOR
234
FOURTH OUTER CONDUCTOR
154
FOURTH COIL
168
SIXTH INDUCTOR - SECOND TWISTED LOOP CONDUCTOR
156, 238
SEVENTH POLYGONAL CONDUCTOR
252
EIGHTH POLYGONAL CONDUCTOR
254
140
138
L5
L4
L6
256
242
244
126
136

FIG. 7



270
LEGEND
FIRST CIRCUIT LAYER
SECOND CIRCUIT LAYER
172
174
120, 146, 148, 150, 162, 164, 194, 196, 198, 212, 214
272
122, 152, 154, 156, 166, 168, 236, 252, 254
124
126
130
132
134
136
138
140
158
190
192
216
218
256
L1
L2
L3
L4
L5
L6

FIG. 8

COMPACT PHASE SHIFTER LAYOUT

BACKGROUND

[0001] The present disclosure relates generally to phase shifter circuitry of an electronic device.

[0002] Phase shifter circuitry may adjust a phase of input signals by a desired value. For example, an electronic device may include the phase shifter circuitry. The electronic device may generate transmission signals in the form of a beam with desired beam characteristics. The electronic device may adjust a phase of the transmission signals by the phase shifter circuitry based on a direction of the desired beam. The electronic device may then transmit the transmission signals as adjusted. Similarly, the electronic device may receive signals having desired beam characteristics. The electronic device may adjust a phase of the received signals by the phase shifter circuitry to receive the signals.

[0003] Restrictions in a layout of the phase shifter circuitry may result in reducing signal quality, decreased area occupied by the phase shifter circuitry, among other things. Improved phase shifter circuitry layouts are desired to improve the signal quality and/or reduce an area occupied by the phase shifter circuitry.

[0004] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

SUMMARY

[0005] A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

[0006] In one embodiment, phase shifter circuitry may include a first phase shifter circuit including a first coil and a second coil. The second coil may be disposed over and extended around a boundary of the first coil, and the second coil may include a first twisted loop conductor extended in a first direction. The phase shifter circuitry may also include a second phase shifter circuit including a third coil and a fourth coil. The fourth coil may be disposed over and enclosed by a boundary of the third coil, and the fourth coil may include a second twisted loop conductor extended in a second direction different than the first direction.

[0007] In another embodiment, phase shifter circuitry may include a first coil disposed on a first circuit layer and having a first boundary. The first coil may include a first polygonal conductor including a first inner conductor coupled to a second inner conductor. The first coil may also include a second polygonal conductor including a first outer conductor coupled to a second outer conductor. The first outer conductor may be coupled to the first inner conductor, the second outer conductor may be coupled to the second inner conductor, and the first polygonal conductor may be encircled by the second outer conductor. The phase shifter circuitry may also include a second coil disposed on a second circuit layer. The second coil may be coupled to the first inner conductor and the second inner conductor, and the second coil may include a first twisted loop conductor having a second boundary that covers the first boundary of the first coil. Additionally, the phase shifter circuitry may include a third coil disposed on the first circuit layer. The third coil may be coupled to the second outer conductor, and the third coil may not overlap with the first coil and the second coil. Further, the phase shifter circuitry may include a fourth coil coupled to the third coil. The fourth coil may include a second twisted loop conductor encircled by the third coil, and the second twisted loop conductor may be disposed in a different direction with respect to the first twisted loop conductor.

[0008] In yet another embodiment, an electronic device includes processing circuitry, an antenna, and phase shifter circuitry coupled to the antenna and the processing circuitry. The phase shifter circuitry may include a first coil and a second coil coupled to the first coil. The second coil may be disposed over and extend around the first coil, and the second coil may include a first twisted loop conductor extended in a first direction. The phase shifter circuitry may also include a third coil coupled to the first coil. The third coil may not overlap with the first coil and the second coil. Further, the phase shifter circuitry may include a fourth coil coupled to the third coil. The fourth coil may be disposed over and surrounded by the third coil, and the fourth coil may include a second twisted loop conductor extending in a second direction different than the first direction.

[0009] Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended only to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings in which:

[0011] FIG. **1** is a block diagram of an electronic device, according to embodiments of the present disclosure;

[0012] FIG. **2** is a functional diagram of the electronic device of FIG. **1**, according to embodiments of the present disclosure;

[0013] FIG. **3** is a schematic diagram of a transmitter of the electronic device of FIG. **1**, according to embodiments of the present disclosure;

[0014] FIG. **4** is a schematic diagram of a receiver of the electronic device of FIG. **1**, according to embodiments of the present disclosure;

[0015] FIG. **5** is a schematic diagram of phase shifter circuitry of the electronic device of FIGS. **1-4**, according to embodiments of the present disclosure;

[0016] FIG. **6** is a layout of a first phase shifter circuit of the phase shifter circuitry of FIG. **5**, according to embodiments of the present disclosure;

[0017] FIG. **7** is a layout of a second phase shifter circuit of the phase shifter circuitry of FIG. **5**, according to embodiments of the present disclosure; and

[0018] FIG. **8** is a layout of the phase shifter circuitry of FIGS. **6** and **7** including the first phase shifter circuit and the second phase shifter circuit, according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

[0019] When introducing elements of various embodiments of the present disclosure, the articles "a," "an," and "the" are intended to mean that there are one or more of the elements. The terms "comprising," "including," and "having" are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to "one embodiment" or "an embodiment" of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Use of the terms "approximately," "near," "about," "close to," and/or "substantially" should be understood to mean including close to a target (e.g., design, value, amount), such as within a margin of any suitable or contemplatable error (e.g., within 0.1% of a target, within 1% of a target, within 5% of a target, within 10% of a target, within 25% of a target, and so on). Moreover, it should be understood that any exact values, numbers, measurements, and so on, provided herein, are contemplated to include approximations (e.g., within a margin of suitable or contemplatable error) of the exact values, numbers, measurements, and so on. Additionally, the term "set" may include one or more. That is, a set may include a unitary set of one member, but the set may also include a set of multiple members.

[0020] This disclosure is directed to phase shifter circuitry with reduced insertion loss and/or reduced area compared to other phase shifters. The phase shifter circuitry may include, for example, a first phase shifter circuit and a second phase shifter circuit. In some embodiments, the first phase shifter circuit and the second phase shifter circuit may each include two coils forming three inductors.

[0021] The first phase shifter circuit and the second phase shifter circuit may each reduce induced currents of the phase shifter circuitry during operation. The first phase shifter circuit and the second phase shifter circuit may have non-overlapping boundaries. That is, the first phase shifter circuit and the second phase shifter circuit may each direct input signals through a number of polygonal conductors that are disposed adjacently or overlaid along multiple planes of a circuit board. The polygonal conductors of each of the first phase shifter circuit and the second phase shifter circuit may be non-overlapping or at least partially overlap with one another. The planes of the circuit board may be disposed in parallel to one another. In some cases, the induced currents of one or more of the polygonal conductors destructively combine to reduce induced currents of the phase shifter circuitry. As such, the phase shifter circuitry may output signals with improved linearity based on the reduced induced currents.

[0022] Moreover, the first phase shifter circuit and the second phase shifter circuit may each include orthogonally disposed figure-eight shaped polygonal conductors that are non-overlapping. In some cases, the orthogonal and non-overlapping disposition of the figure-eight shaped polygonal conductors may increase a distance between the polygonal conductors of the first phase shifter circuit and the second phase shifter circuit. The increased distance between the polygonal conductors may reduce the induced currents of the phase shifter circuitry.

[0023] In some embodiments, the phase shifter circuitry may have reduced insertion loss based on the reduction of the induced currents. In some cases, the phase shifter circuitry may output signals with improved linearity based on the reduced induced currents. Moreover, in specific embodiments, the phase shifter circuitry may have a reduced area compared to other phase shifters based on the disposition of the polygonal conductors of each of the inductors.

[0024] FIG. **1** is a block diagram of an electronic device **10**, according to embodiments of the present disclosure. The electronic device **10** may include, among other things, one or more processors **12** (collectively referred to herein as a single processor for convenience, which may be implemented in any suitable form of processing circuitry), memory **14**, nonvolatile storage **16**, a display **18**, input structures **22**, an input/output (I/O) interface **24**, a network interface **26**, and a power source **29**. The various functional blocks shown in FIG. **1** may include hardware elements (including circuitry), software elements (including machine-executable instructions) or a combination of both hardware and software elements (which may be referred to as logic). The processor **12**, memory **14**, the nonvolatile storage **16**, the display **18**, the input structures **22**, the input/output (I/O) interface **24**, the network interface **26**, and/or the power source **29** may each be communicatively coupled directly or indirectly (e.g., through or via another component, a communication bus, a network) to one another to transmit and/or receive signals between one another. It should be noted that FIG. **1** is merely one example of a particular implementation and is intended to illustrate the types of components that may be present in the electronic device **10**.

[0025] By way of example, the electronic device **10** may include any suitable computing device, including a desktop or notebook computer, a portable electronic or handheld electronic device such as a wireless electronic device or smartphone, a tablet, a wearable electronic device, and other similar devices. In additional or alternative embodiments, the electronic device **10** may include an access point, such as a base station, a router (e.g., a wireless or Wi-Fi router), a hub, a switch, and so on. It should be noted that the processor **12** and other related items in FIG. **1** may be embodied wholly or in part as software, hardware, or both. Furthermore, the processor **12** and other related items in FIG. **1** may be a single contained processing module or may be incorporated wholly or partially within any of the other elements within the electronic device **10**. The processor **12** may be implemented with any combination of general-purpose microprocessors, microcontrollers, digital signal processors (DSPs), field programmable gate array (FPGAs), programmable logic devices (PLDs), controllers, state machines, gated logic, discrete hardware components, dedicated hardware finite state machines, or any other suitable entities that may perform calculations or other manipulations of information. The processors **12** may include one or more application processors, one or more baseband processors, or both, and perform the various functions described herein.

[0026] In the electronic device **10** of FIG. **1**, the processor **12** may be operably coupled with a memory **14** and a nonvolatile storage **16** to perform various algorithms. Such programs or instructions executed by the processor **12** may be stored in any suitable article of manufacture that includes one or more tangible, computer-readable media. The tangible, computer-readable media may include the memory **14** and/or the nonvolatile storage **16**, individually or collectively, to store the instructions or routines. The memory **14** and the nonvolatile storage **16** may include any suitable articles of manufacture for storing data and executable instructions, such as random-access memory, read-only memory, rewritable flash memory, hard drives, and optical discs. In addition, programs (e.g., an operating system) encoded on such a computer program product may also include instructions that may be executed by the processor **12** to enable the electronic device **10** to provide various functionalities.

[0027] In certain embodiments, the display **18** may facilitate users to view images generated on the electronic device **10**. In some embodiments, the display **18** may include a touch screen, which may facilitate user interaction with a user interface of the electronic device **10**. Furthermore, it should be appreciated that, in some embodiments, the display **18** may include one or more liquid crystal displays (LCDs), light-emitting diode (LED) displays, organic light-emitting diode (OLED) displays, active-matrix organic light-emitting diode (AMOLED) displays, or some combination of these and/or other display technologies.

[0028] The input structures **22** of the electronic device **10** may enable a user to interact with the electronic device **10** (e.g., pressing a button to increase or decrease a volume level). The I/O interface **24** may enable electronic device **10** to interface with various other electronic devices, as may the network interface **26**. In some embodiments, the I/O interface **24** may include an I/O port for a hardwired connection for charging and/or content manipulation using a standard connector and protocol, such as the Lightning connector, a universal serial bus (USB), or other similar connector and protocol. The network interface **26** may include, for example, one or more interfaces for a personal area network (PAN), such as an ultra-wideband (UWB) or a BLUETOOTH® network, a local area network (LAN) or wireless local area network (WLAN), such as a network employing one of the IEEE 802.11x family of protocols (e.g., WI-FI®), and/or a wide area network (WAN), such as any standards related to the Third Generation Partnership Project (3GPP), including, for example, a 3rd generation (3G) cellular network, universal mobile telecommunication system (UMTS), 4th generation (4G) cellular network, Long Term Evolution® (LTE) cellular network, Long Term Evolution License Assisted Access (LTE-LAA) cellular network, 5th generation (5G) cellular network, and/or New Radio (NR) cellular network, a 6th generation (6G) or greater than 6G cellular network, a satellite network, a non-terrestrial network, and so on. In particular, the network interface **26** may include, for example, one or more interfaces for using a cellular communication standard of the 5G specifications that include the millimeter wave (mmWave) frequency range (e.g., 24.25-300 gigahertz (GHz)) that defines and/or enables frequency ranges used for wireless communication. The network interface **26** of the electronic device **10** may allow communication over the aforementioned networks (e.g., 5G, Wi-Fi, LTE-LAA, and so forth).

[0029] The network interface **26** may also include one or more interfaces for, for example, broadband fixed wireless access networks (e.g., WIMAX®), mobile broadband Wireless networks (mobile WIMAX®), asynchronous digital subscriber lines (e.g., ADSL, VDSL), digital video broadcasting-terrestrial (DVB-T®) network and its extension DVB Handheld (DVB-H®) network, ultra-wideband (UWB) network, alternating current (AC) power lines, and so forth. The power source **29** of the electronic device **10** may include any suitable source of power, such as a rechargeable lithium polymer (Li-poly) battery and/or an alternating current (AC) power converter.

[0030] As illustrated, the network interface **26** may include a transceiver **30**. In some embodiments, all or portions of the transceiver **30** may be disposed within the processor **12**. The transceiver **30** may support transmission and receipt of various wireless signals via one or more antennas, and thus may include a transmitter and a receiver. In some embodiments, the transceiver **30** may include phase shifter circuitry. The phase shifter circuitry may include, for example, a first phase shifter circuit and a second phase shifter circuit having non-overlapping boundaries. In some embodiments, the first phase shifter circuit and the second phase shifter circuit may each include two coils forming three inductors. Moreover, the first phase shifter circuit and the second phase shifter circuit may each direct input signals through a number of polygonal conductors of the respective coils that are disposed adjacently or overlaid along multiple planes of a circuit board.

[0031] FIG. **2** is a functional diagram of the electronic device **10** of FIG. **1**, according to embodiments of the present disclosure. As illustrated, the processor **12**, the memory **14**, the transceiver **30**, a transmitter **52**, a receiver **54**, and/or antennas **55** (illustrated as **55**A-**55**N, collectively referred to as an antenna **55**) may be communicatively coupled directly or indirectly (e.g., through or via another component, a communication bus, a network) to one another to transmit and/or receive signals between one another.

[0032] The electronic device **10** may include the transmitter **52** and/or the receiver **54** that respectively enable transmission and reception of signals between the electronic device **10** and an external device via, for example, a network (e.g., including base stations or access points) or a direct connection. As illustrated, the transmitter **52** and the receiver **54** may be combined into the transceiver **30**. In some embodiments, the transmitter **52**, the receiver **54**, or both, may include phase shifter circuitry. In some cases, the phase shifter circuitry may adjust a phase of transmission signals for transmission by antennas **55**A-**55**N. For example, the phase shifter circuitry may adjust a phase of the transmission signals based on a direction of a desired transmission beam. Moreover, the phase shifter circuitry may adjust a phase of received signals by antennas **55**A-**55**N for reception by the processor **12** and/or the memory **14**, among other things. For example, the phase shifter circuitry may adjust a phase of the received signals based on a direction of a reception beam associated with the received signals.

[0033] As mentioned above, the phase shifter circuitry may include, for example, a first phase shifter circuit and a second phase shifter circuit having non-overlapping boundaries. The first phase shifter circuit and the second phase shifter circuit may each reduce induced currents of the phase shifter circuitry during operation. In some embodiments, the phase shifter circuitry may have reduced insertion loss and/or improved linearity. In some embodiments, the phase shifter circuitry may adjust a phase of the transmission signals and/or reception signals with improved accuracy compared to other phase shifter. Alternatively or additionally, the phase shifter circuitry may have a reduced area compared to other phase shifters, as will be appreciated.

[0034] The electronic device **10** may also have the antennas **55**A-**55**N electrically coupled to the transceiver **30**. The antennas **55**A-**55**N may be configured in an omnidirectional or directional configuration, in a single-beam, dual-beam, or multi-beam arrangement, and so on. Each antenna **55** may be associated with one or more beams and various configurations. In some embodiments, multiple antennas of the antennas **55**A-**55**N of an antenna group or module may be communicatively coupled to a respective transceiver **30** and each emit radio frequency signals that may constructively and/or destructively combine to form a beam. The electronic device **10** may include multiple transmitters, multiple receivers, multiple transceivers, and/or multiple antennas as suitable for various communication standards. In some embodiments, the transmitter **52** and the receiver **54** may transmit and receive information via other wired or wireline systems or means.

[0035] As illustrated, the various components of the electronic device **10** may be coupled together by a bus system **56**. The bus system **56** may include a data bus, for example, as well as a power bus, a control signal bus, and a status signal bus, in addition to the data bus. The components of the electronic device **10** may be coupled together or accept or provide inputs to each other using some other mechanism.

[0036] FIG. **3** is a schematic diagram of the transmitter **52** (e.g., transmit circuitry) of the transceiver **30**, according to embodiments of the present disclosure. As illustrated, the transmitter **52** may receive outgoing data **60** in the form of a digital signal to be transmitted via the one or more antennas **55**. In some embodiments, the transmitter **52** may receive the outgoing data **60** from the processor **12**. A digital-to-analog converter (DAC) **62** of the transmitter **52** may convert the digital signal to an analog signal, and a modulator **64** may combine the converted analog signal with a carrier signal to generate a radio wave **67** (e.g., a modulated signal).

[0037] In some embodiments, a power amplifier (PA) **66** may be coupled to an output of the modulator **64**. In the depicted embodiment, phase shifter circuitry **68** may be coupled to an output of the modulator **64**. In some cases, the phase shifter circuitry **68** may adjust a phase of the radio wave **67**. For example, the phase shifter circuitry **68** may adjust a phase of the radio wave **67** based on a desired direction or characteristics of a transmission beam. That is, the phase of the radio wave **67** may correspond to a direction of the transmission beam. As such, adjusting the phase of the radio wave **67** may adjust the direction of the transmission beam.

[0038] The power amplifier **66** receives the phase shifted radio wave **67** (e.g., the phase shifted and modulated signal) from the phase shifter circuitry **68**. The power amplifier **66** may amplify the radio wave **67** to generate amplified signals. The power amplifier **66** may generate the amplified signal to a suitable level to drive transmission of the signal via the one or more antennas **55**. It should be appreciated that in alternative or additional embodiments, the phase power amplifier **66** may be disposed before the shifter circuitry **68** and/or after a filter **69**, among other possibilities.

[0039] The filter **69** (e.g., filter circuitry and/or software) of the transmitter **52** may then remove undesirable noise from the phase shifted amplified radio wave **67** to generate transmitted signal **70** to be transmitted via the one or more antennas **55**. The filter **69** may include any suitable filter or filters to remove the undesirable noise from the amplified signal, such as a bandpass filter, a bandstop filter, a low pass filter, a high pass filter, and/or a decimation filter.

[0040] The power amplifier **66** and/or the filter **69** may be referred to as part of a radio frequency front end (RFFE), and more specifically, a transmit front end (TXFE) of the electronic device **10**. Additionally, the transmitter **52** may include any suitable additional components not shown, or may not include certain of the illustrated components, such that the transmitter **52** may transmit the outgoing data **60** via the one or more antennas **55**. For example, the transmitter **52** may include a mixer and/or a digital up converter. As another example, the transmitter **52** may not include the filter **69** if the power amplifier **66** outputs the amplified signal in or approximately in a desired frequency range (such that filtering of the amplified signal may be unnecessary). It should be appreciated that the transceiver **30** may include multiple branches of the transmitter **52**, each including similar or different components.

[0041] FIG. **4** is a schematic diagram of the receiver **54** (e.g., receive circuitry), according to embodiments of the present disclosure. As illustrated, the receiver **54** may receive received signal **80** from the one or more antennas **55** in the form of an analog signal. A low noise amplifier (LNA) **82** may amplify the received analog signal to generate an amplified received signal **83**. In some embodiments, a filter may be coupled to an input of the low noise amplifier (LNA) **82**. The amplified received signal **83** may amplify the received analog signal to a suitable level for the receiver **54** to process.

[0042] In some cases, the phase shifter circuitry **68** may adjust a phase of the amplified received signal **83**. For example, the phase shifter circuitry **68** may adjust a phase of the amplified received signal **83** based on a desired direction or characteristics of a received beam. That is, the phase of the amplified received signal **83** may correspond to a direction of the received beam. As such, adjusting the phase of the amplified received signal **83** may adjust the direction or characteristics of the received beam (e.g., used to receive the received signal **80**). It should be appreciated that in alternative or additional embodiments, the phase shifter circuitry **68** may be disposed before the low noise amplifier **82** and/or after a filter **84**, among other possibilities.

[0043] In some embodiments, the transmitter **52** discussed above and the receiver **54** may include (e.g., share) the phase shifter circuitry **68**. Alternatively or additionally, the transmitter **52** may include first phase shifter circuitry **68** and the receiver **54** may include second phase shifter circuitry **68**. For example, in some embodiments, the first phase shifter circuitry **68** and the second phase shifter circuitry **68** may include similar circuitry and components.

[0044] The filter **84** (e.g., filter circuitry and/or software) may remove undesired noise from the amplified received signal **83**, such as cross-channel interference. The filter **84** may also remove additional signals received by the one or more antennas **55** that are at frequencies other than the desired signal. The filter **84** may include any suitable filter or filters to remove the undesired noise or signals from the received signal, such as a bandpass filter, a bandstop filter, a low pass filter, a high pass filter, and/or a decimation filter. The low noise amplifier **82** and/or the filter **84** may be referred to as part of the RFFE, and more specifically, a receiver front end (RXFE) of the electronic device **10**.

[0045] A demodulator **86** may remove a radio frequency carrier signal and/or extract a demodulated signal (e.g., an envelope signal) from the filtered signal for processing. An analog-to-digital converter (ADC) **88** may receive the demodulated analog signal and convert the signal to a digital signal of incoming data **90** to be further processed by the electronic device **10**. Additionally, the receiver **54** may include any suitable additional components not shown, or may not include certain of the illustrated components, such that the receiver **54** may receive the received signal **80** via the one or more antennas **55**. For example, the receiver **54** may include a mixer and/or a digital down converter. It should be appreciated that the transceiver **30** may include multiple branches of the receiver **54**, each including similar or different components.

[0046] FIG. **5** is a schematic diagram of the phase shifter circuitry **68**, according to embodiments of the present disclosure. As mentioned above, the transceiver **30** may include the phase shifter circuitry **68**. The phase shifter circuitry **68** may include a first phase shifter circuit **120** and a second phase shifter circuit **122**. The first phase shifter circuit **120** may shift (e.g., delay) the phase of input signals by a first phase shift value (e.g., −25° or less, −47 or less, −65° or less, −90° or less, −98° or less, −90° or more, and so on, among other possibilities). Moreover, the second phase shifter circuit **122** may shift (e.g., delay) the phase of the input signals by a second phase shift value (e.g., −25° or less, −32° or less, −39° or less, −45° or less, −52° or less, −45° or more, and so on, among other possibilities).

[0047] The first phase shifter circuit **120** may have a first terminal **124**. The first phase shifter circuit **120** may be coupled to the second phase shifter circuit **122**. Moreover, the second phase shifter circuit **122** may have a second terminal **126**. In some embodiments, the phase shifter circuitry **68** may receive the input signals from the first terminal **124**. In alternative or additional embodiments, the phase shifter circuitry **68** may receive the input signals from the second terminal **126**. The input signals may include the radio wave **67**, the amplified signal, and/or the amplified received signal **83** discussed above, among other possibilities. Similarly, the first terminal **124** and the second terminal **126** may output signals by the first terminal **124** and/or the second terminal **126**. The output signals may include the radio wave **67**, the amplified signal, and/or the amplified received signal **83**, as adjusted.

[0048] The first phase shifter circuit **120** may include a first switch **130**, a second switch **132**, and a third switch **134**. The first phase shifter circuit **120** may include a first inductor **146** (L1) and a second inductor **148** (L2) coupled to the first switch **130** and the second switch **132**. The first inductor **146** may be coupled to the first terminal **124**. In the depicted embodiment, the second inductor **148** may be coupled to the first switch **130** via a capacitor **158**. The first inductor **146** and the second inductor **148** may have different or opposite polarities. Moreover, the first inductor **146** and the second inductor **148** may be coupled in parallel when the first switch **130** is closed.

[0049] The first inductor **146** and the second inductor **148** may be coupled to a third inductor **150** (L3) via the second switch **132**. The third inductor **150** may be coupled to the second switch **132** and the third switch **134** on one end and to a ground connection on a second end. The third inductor **150** may be coupled to the first inductor **146** and the second inductor **148** via the second switch **132** and may be coupled to the ground connection via the third switch **134**. The third switch **134** may close (e.g., activate) and/or the second switch may open (e.g., deactivate) to couple the third inductor **150** to the ground connection and therefore bypass the third inductor **150**.

[0050] The first switch **130** may be open, the second switch **132** may be open, and the third switch **134** may be closed to adjust a phase of the input signals by the first phase shift value. The first switch **130** may be closed, the second switch **132** may be closed, and the third switch **134** may be open to bypass the first phase shifter circuit **120**. As such, the first phase shifter circuit **120** may be coupled to the second phase shifter circuit **122** via the first switch **130** and the capacitor **158**.

[0051] Moreover, the second phase shifter circuit **122** may include a fourth switch **136**, a fifth switch **138**, and a sixth switch **140**. The second phase shifter circuit **122** may include a fourth inductor **152** (L4) and a fifth inductor **154** (L5) coupled to the fourth switch **136** and the fifth switch **138**. The fifth inductor **154** may be coupled to the second terminal **126**. The fourth inductor **152** and the fifth inductor **154** may have different or opposite polarities. Moreover, the fourth inductor **152** and the fifth inductor **154** may be coupled in parallel when the fourth switch **136** is closed.

[0052] The fourth inductor **152** and the fifth inductor **154** may be coupled to a sixth inductor **156** (L6) via the fifth switch **138**. The sixth inductor **156** may be coupled to the fifth switch **138** and the sixth switch **140** on one end and to the ground connection on a second end. The sixth inductor **156** may be coupled to the first inductor **146** and the second inductor **148** via the fifth switch **138** and may be coupled to the ground connection via the sixth switch **140**. The sixth switch **140** may close (e.g., activate) and/or the fifth switch may open (e.g., deactivate) to couple the sixth inductor **156** to the ground connection and therefore bypass the sixth inductor **156**.

[0053] The fourth switch **136** may be open, the fifth switch **138** may be open, and the sixth switch **140** may be closed to adjust a phase of the input signals by the second phase shift value. The fourth switch **136** may be closed, the fifth switch **138** may be closed, and the sixth switch **140** may be open to bypass the second phase shifter circuit **122**. As such, the phase shifter circuitry **68** may adjust a phase of the input signals by the first phase shift value, the second phase shift value, or both. The processor **12** discussed above, or any other viable component may generate control signals to open and close the switches **130**, **132**, **134**, **136**, **138**, and **140**.

[0054] FIG. **6** is a layout **160** of the first phase shifter circuit **120** of the phase shifter circuitry **68**, according to embodiments of the present disclosure. The first phase shifter circuit **120** may include a first coil **162** and a second coil **164**. The first coil **162** may form the first inductor **146**

(L1) and the second inductor **148** (L2). Moreover, the second coil **164** may form the third inductor **150** (L3), as will be appreciated. As mentioned above, the first phase shifter circuit **120** may shift (e.g., delay) the phase of the input signals by a first phase shift value (e.g., −25°, −47, −65°, −90°, −98°, and so, among other possibilities).

[0055] The first coil **162** may be disposed on a first circuit layer **172** (e.g., a first plane surface) of the phase shifter circuitry **68**. The second coil **164** may be disposed on a second circuit layer **174** (e.g., a second plane surface) of the phase shifter circuitry **68**. The first circuit layer **172** may be disposed over or under, and in proximity of (e.g., adjacent to) the second circuit layer **174**. For example, the first circuit layer **172** and the second circuit layer **174** may each be disposed on different circuit layers of a printed circuit board (PCB), among other possibilities.

[0056] The first coil **162** may include a first outer conductor **180**, a first inner conductor **182**, a second inner conductor **184**, and a second outer conductor **186**. The conductors **180**, **182**, **184**, and **186** may each form a portion of a respective polygonal shape, as discussed herein. The first outer conductor **180** may be coupled to the first terminal **124** of the phase shifter circuitry **68**. Moreover, the second outer conductor **186** may be coupled to the second phase shifter circuitry **68**. As mentioned above, in some embodiments, the first phase shifter circuit **120** may receive input signals from the first terminal **124**. In alternative or additional embodiments, the first phase shifter circuit **120** may receive input signals from the second phase shifter circuitry **68**.

[0057] In some embodiments, the first outer conductor **180** may be disposed at least partially symmetrical to the second outer conductor **186**. The first outer conductor **180** may be disposed in proximity of (e.g., adjacent to) and around the second inner conductor **184** on the first circuit layer **172**. That is, the first outer conductor **180** may be disposed circularly outside of (e.g., around) and partially surrounding the second inner conductor **184**.

[0058] Similarly, the second outer conductor **186** may be disposed in proximity of (e.g., adjacent to) and around the first inner conductor **182** on the first circuit layer **172**. That is, the second outer conductor **186** may be disposed concentrically outside of (e.g., around) and partially surrounding the first inner conductor **182**. The first inner conductor **182** and the second inner conductor **184** may be coupled to and disposed between the first outer conductor **180** and the second outer conductor **186**.

[0059] As such, the first outer conductor **180** may be coupled to the second outer conductor **186** via the first inner conductor **182** and the second inner conductor **184**. In some embodiments, the first outer conductor **180** may also be coupled to the second outer conductor **186** via the first switch **130** and the capacitor **158**. As mentioned above, the first switch **130** may close to bypass the phase shifter circuitry **68** by coupling the first terminal **124** to the second phase shifter circuitry **68**.

[0060] The first inner conductor **182** may be coupled to the first outer conductor **180** by a first connector **190**. The second inner conductor **184** may be coupled to the second outer conductor by a second connector **192** cross-coupled over or under the first connector **190**. Moreover, the first inner conductor **182** may be coupled to the second inner conductor **184** to form a first polygonal conductor **194**. As such, the first polygonal conductor **194** may be disposed between and/or twisted inside the first outer conductor **180** and the second outer conductor **186**.

[0061] The first coil **162** may include the first connector **190**, the second connector **192**, or both. The first connector **190** may be disposed (e.g., crossed) over or under the second connector **192** across multiple circuit layers to cross-couple the outer conductors **180** and **186** to the inner conductors **182** and **184**. In specific embodiments, the first connector **190** and the second connector **192** may each be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities.

[0062] In the depicted embodiment, the first polygonal conductor **194** may be enclosed (e.g., encircled) by the first outer conductor **180** and the second outer conductor **186**. A polygonal conductor, such as the first polygonal conductor **194**, may have any viable polygonal shape and/or symmetrical polygonal shape, among other things. For example, the first polygonal conductor **194**, including the first inner conductor **182** and the second inner conductor **184**, may have a circular shape, a pentagonal shape, an octagonal shape, a hexagonal shape, among other possibilities.

[0063] In some embodiments, the first outer conductor **180** and the second outer conductor **186** may be symmetrical or at least partially symmetrical. Moreover, the first outer conductor **180** may be disposed around the second inner conductor **184** and the second outer conductor **186** may be disposed around the first inner conductor **182**. In some embodiments, the first outer conductor **180** and the first inner conductor **182** may form at least a part of a second polygonal conductor **196** (e.g., a symmetrical polygonal conductor). The first inductor **146** may include the second polygonal conductor **196** including the first outer conductor **180** and the first inner conductor **182**.

[0064] Similarly, the second outer conductor **186** and the second inner conductor **184** may form at least a part of a third polygonal conductor **198** (e.g., a symmetrical polygonal conductor). The second inductor **148** may include the third polygonal conductor **198** including the second outer conductor **186** and the second inner conductor **184**. In the depicted embodiment, a first area associated with the first inductor **146** may at least partially overlap with a second area associated with the second inductor **148**. The first area may be partially enclosed by the second polygonal conductor **196**. Moreover, the second area may be partially enclosed by the third polygonal conductor **198**. As such, the first inductor **146** and the second inductor **148** may be at least partially intertwined.

[0065] The first inductor **146** may be coupled to the first terminal **124** and the second inductor **148**. Moreover, the second inductor **148** may be coupled to the first inductor **146** and the second phase shifter circuit **122**. As mentioned above, the first inductor **146** of the first phase shifter circuit **120** may receive input signals from the first terminal **124** and provide output signals to the second phase shifter circuitry **68**. Alternatively or additionally, the second inductor **148** of the first phase shifter circuit **120** may receive the input signals from the second phase shifter circuitry **68** and provide the output signals to the first terminal **124**.

[0066] The second coil **164** may include a first twisted loop conductor **210** having a figure-eight shape. The first twisted loop conductor **210** may include a fourth polygonal conductor **212** (e.g., a symmetrical polygonal conductor) and a fifth polygonal conductor **214** (e.g., a symmetrical polygonal conductor). The fourth polygonal conductor **212**

may be cross-coupled to the fifth polygonal conductor **214** (e.g., a symmetrical polygonal conductor).

[0067] The fourth polygonal conductor **212** and the fifth polygonal conductor **214** may each have a circular shape, a pentagonal shape, an octagonal shape, and/or a hexagonal shape, among other possibilities. The third inductor **150** may include the fourth polygonal conductor **212** and the fifth polygonal conductor **214**.

[0068] The fourth polygonal conductor **212** may be cross-coupled to the fifth polygonal conductor **214** by a third connector **216** and a fourth connector **218**. The second coil **164** may include the third connector **216**, the fourth connector **218**, or both. The third connector **216** may be disposed (e.g., crossed) over or under the fourth connector **218** across multiple circuit layers to cross-couple the fourth polygonal conductor **212** to the fifth polygonal conductor **214**. In specific embodiments, the third connector **216** and the fourth connector **218** may each be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities. As such, the second coil **164** may form the first twisted loop conductor **210** based on cross-coupling the fourth polygonal conductor **212** to the fifth polygonal conductor **214**.

[0069] The second coil **164** may have a first input port and a first output port on the fourth polygonal conductor **212**. The first output port of the second coil **164** may be coupled to a ground connection. The first input port may be coupled to an intersection of the first inner conductor **182** and the second inner conductor **184** of the first coil **162**. In specific embodiments, the first inner conductor **182** may be split from the second inner conductor **184** at a coupling point of the first input port.

[0070] In some embodiments, the first input port may be coupled to the intersection of the first inner conductor **182** and the second inner conductor **184** via a second switch **132**. The second switch **132** may couple and uncouple the first coil **162** and the second coil **164**. As such, the second switch **132** may open to bypass the second coil **164**. The second switch **132** may be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities. In some embodiments, the first input port may also be coupled to the ground connection via a third switch **134**. As such, the third switch **134** may close to couple the second coil **164** to the ground connection and therefore bypass the second coil **164**.

[0071] With the foregoing in mind, the third inductor **150** may receive at least a portion of the input signals. As mentioned above, the third inductor **150** may receive at least a portion of the input signals based on the first switch **130** being open, the second switch **132** being closed, and the third switch **134** being open. The fourth polygonal conductor **212** and the fifth polygonal conductor **214** may direct the portion of the input signals in opposite directions based on being cross-coupled and having the figure-eight shape.

[0072] In some cases, directing the input signals in the opposite directions through the adjacent polygonal conductors **212** and **214** of the second coil **164** may reduce induced currents of the first phase shifter circuit **120** during operation. For example, in specific cases, at least a part of the induced currents of the polygonal conductors **212** and **214** may destructively combine during operation.

[0073] Moreover, in the depicted embodiment, the second coil **164** may be disposed on the second circuit layer **174** overlaid on and at least partially extended around (e.g., outside) a boundary of the first coil **162** on the first circuit layer **172**. For example, the first twisted loop conductor **210** may have a second boundary that covers (e.g., substantially covers) a first boundary of the first coil **162**. In some cases, the first inductor **146** and the second inductor **148** disposed on the first circuit layer **172** may reduce induced currents of the third inductor **150** disposed on the second circuit layer **174** during operation.

[0074] Similarly, the third inductor **150** may reduce induced currents of the first inductor **146** and the second inductor **148** during operation. For example, in specific cases, at least a part of the induced currents of the inductors **146**, **148**, and **150** may destructively combine during operation. It should be appreciated that in alternative or additional embodiments, the first coil **162** may be overlaid on and at least partially extended around (e.g., outside) the boundary of the second coil **164**.

[0075] FIG. **7** is a layout **230** of the second phase shifter circuit **122** of the phase shifter circuitry **68**, according to embodiments of the present disclosure. The second phase shifter circuit **122** may include a third coil **166** and a fourth coil **168**. As mentioned above, the second phase shifter circuit **122** may shift (e.g., delay) the phase of the input signals by a second phase shift value (e.g., −25°, −32°, −39°, −45°, −52°, and so, among other possibilities). The third coil **166** may form the fourth inductor **152** (L**4**) and the fifth inductor **154** (L**5**). Moreover, the fourth coil **168** may form the sixth inductor **156** (L**6**).

[0076] The third coil **166** may be disposed on the first circuit layer **172** (e.g., the first plane surface) of the phase shifter circuitry **68**. The fourth coil **168** may be disposed on the second circuit layer **174** (e.g., the second plane surface) of the phase shifter circuitry **68**. As mentioned above, the first circuit layer **172** may be disposed over or under, and in proximity of (e.g., adjacent to) the second circuit layer **174**.

[0077] The third coil **166** may include a third outer conductor **232** and a fourth outer conductor **234** forming a sixth polygonal conductor **236** (e.g., a symmetrical polygonal conductor). In some embodiments, the third outer conductor **232** and the fourth outer conductor **234** may each form a respective portion of the sixth polygonal conductor **236**. The third outer conductor **232** may be twisted over or under the fourth outer conductor **234** to form the sixth polygonal conductor **236**. The sixth polygonal conductor **236** may form a circular shape, a pentagonal shape, an octagonal shape, a hexagonal shape, among other possibilities.

[0078] In some cases, the fourth coil **168** may couple to an intersection of the third outer conductor **232** and the fourth outer conductor **234**. The fourth coil **168** may include a second twisted loop conductor **238** disposed between, enclosed by (e.g., encircled by), and/or coupled to the third outer conductor **232** and the fourth outer conductor **234**, as will be appreciated.

[0079] In some embodiments, the third outer conductor **232** may be coupled to the second outer conductor **186** of the first coil discussed above. Moreover, the fourth outer conductor **234** may be coupled to the second terminal **126**. As such, in some cases, the second phase shifter circuit **122** may receive input signals from the second terminal **126**. Alternatively of additionally, the second phase shifter circuit **122** may receive input signals from the first phase shifter circuitry **68**.

[0080] The fourth inductor **152** may include a portion of the sixth polygonal conductor **236** including the third outer conductor **232**. The fifth inductor **154** may include a remaining portion of the sixth polygonal conductor **236** including the fourth outer conductor **234**. In the depicted embodiment, a third area associated with the fourth inductor **152** may at least partially overlap with a fourth area associated with the fifth inductor **154**. The third area may be partially enclosed by the third outer conductor **232**. Moreover, the fourth area may be partially enclosed by the fourth outer conductor **234**. As such, the fourth inductor **152** and the fifth inductor **154** may be at least partially intertwined.

[0081] In some embodiments, the third outer conductor **232** may include a fifth connector **242** and the fourth outer conductor **234** may include a sixth connector **244**. For example, the fifth connector **242** may be disposed (e.g., crossed) over or under the sixth connector **244** across multiple circuit layers to twist the third outer conductor **232** over or under the fourth outer conductor **234**. In specific embodiments, the fifth connector **242** and the sixth connector **244** may each be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities.

[0082] In alternative or additional embodiments, the third outer conductor **232** may be coupled to the fourth outer conductor **234** via a fourth switch **136**. The fourth switch **136** may couple and uncouple an input/output portion of the third outer conductor **232** to an input/output portion of the fourth outer conductor **234**. For example, the fourth switch **136** may couple and uncouple the second outer conductor **186** of the first coil discussed above to the second terminal **126**. As such, the fourth switch **136** may open to bypass the third coil **166**. The fourth switch **136** may be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities.

[0083] The fourth coil **168** may include a second twisted loop conductor **238** having a figure-eight shape. The second twisted loop conductor **238** may include a seventh polygonal conductor **252** (e.g., a symmetrical polygonal conductor) cross-coupled to an eighth polygonal conductor **254** (e.g., a symmetrical polygonal conductor). For example, the seventh polygonal conductor **252** and the eighth polygonal conductor **254** may each have a circular shape, a pentagonal shape, an octagonal shape, and/or a hexagonal shape, among other possibilities. The sixth inductor **156** may include the seventh polygonal conductor **252** and the eighth polygonal conductor **254**.

[0084] In some embodiments, the seventh polygonal conductor **252** and the eighth polygonal conductor **254** may share a portion of the respective conductors. Moreover, the seventh polygonal conductor **252** may be cross-coupled to the eighth polygonal conductor **254** by a seventh connector **256**. In some embodiments, the fourth coil **168** may include the seventh connector **256**. The seventh connector **256** may be disposed (e.g., crossed) over or under a portion of the seventh polygonal conductor **252** and the eighth polygonal conductor **254**.

[0085] The seventh connector **256** may be disposed across multiple circuit layers to cross-couple the seventh polygonal conductor **252** to the eighth polygonal conductor **254**. In specific embodiments, the seventh connector **256** may be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities. As such, the fourth coil **168** may form the second twisted loop conductor **238** based on cross-coupling the seventh polygonal conductor **252** to the eighth polygonal conductor **254**.

[0086] The seventh polygonal conductor **252** may have a second input port and a second output port. The second output port may be coupled to a ground connection. The second input port may be coupled to the intersection of the third outer conductor **232** and the fourth outer conductor **234** of the third coil **166**. In some embodiments, the third outer conductor **232** may be split from the fourth outer conductor **234** at a coupling point of the second input port.

[0087] In some embodiments, the second input port may be coupled to the intersection of the third outer conductor **232** and the fourth outer conductor **234** via a fifth switch **138**. The fifth switch **138** may couple and uncouple the third coil **166** and the fourth coil **168**. The fifth switch **138** may open to bypass the fourth coil **168**. The fifth switch **138** may be disposed on and/or in-between the first circuit layer **172** and/or the second circuit layer **174**, among other possibilities. In some embodiments, the second input port may also be coupled to the ground connection via a sixth switch **140**. As such, the sixth switch **140** may close to couple the fourth coil **168** to the ground connection and therefore bypass the fourth coil **168**.

[0088] The third outer conductor **232** and the fourth outer conductor **234** may be disposed in proximity of (e.g., adjacent to) and around the second twisted loop conductor **238**. The third outer conductor **232** and the fourth outer conductor **234** may be disposed circularly outside of and partially surrounding the second twisted loop conductor **238**. For example, the third coil **166** may be extended around boundaries of the second twisted loop conductor **238**. It should be appreciated that in alternative or additional embodiments, the second twisted loop conductor **238** may be disposed circularly outside of and partially surrounding the third outer conductor **232** and the fourth outer conductor **234**. For example, the second twisted loop conductor **238** may be extended around boundaries of the third coil **166**.

[0089] With the foregoing in mind, the sixth inductor **156** may receive at least a portion of the input signals. As mentioned above, the sixth inductor **156** may receive at least a portion of the input signals based on the fourth switch **136** being open, the fifth switch **138** being closed, and the sixth switch **140** being open. The seventh polygonal conductor **252** and the eighth polygonal conductor **254** may be cross-coupled and have a figure-eight shape to direct the portion of the input signals in opposite directions.

[0090] In some cases, directing the input signals in the opposite directions through the adjacent polygonal conductors **252** and **254** of the fourth coil **168** may reduce induced currents of the second phase shifter circuit **122** during operation. For example, in some cases, at least a part of the induced currents of the polygonal conductors **252** and **254** may destructively combine during operation.

[0091] Moreover, in the depicted embodiment, the third coil **166** may be disposed on the first circuit layer **172** overlaid on and at least partially extended around (e.g., outside) a boundary of the fourth coil **168** on the second circuit layer **174**. In some cases, the fourth inductor **152** and the fifth inductor **154** disposed on the first circuit layer **172** may reduce induced currents of the sixth inductor **156** disposed on the second circuit layer **174** during operation.

[0092] Similarly, the sixth inductor **156** may reduce induced currents of the fourth inductor **152** and the fifth inductor **154** during operation. For example, in some cases, at least a part of the induced currents of the inductors **152**, **154**, and **156** may destructively combine during operation. It should be appreciated that in alternative or additional embodiments, the fourth coil **168** may be overlaid on and at least partially extended around (e.g., outside) the boundary of the third coil **166**.

[0093] FIG. **8** is a layout **270** of the phase shifter circuitry **68** including the first phase shifter circuit **120** and the second phase shifter circuit **122**, according to embodiments of the present disclosure. The first phase shifter circuit **120** and the second phase shifter circuit **122** may be coupled to a ring guard **272**. As discussed above, the first phase shifter circuit **120** may include a first coil **162** and a second coil **164**. Moreover, the second phase shifter circuit **122** may include a third coil **166** and a fourth coil **168**.

[0094] The third coil **166** and the fourth coil **168** may be disposed adjacent to and not having an overlapping boundary with the first coil **162** and the second coil **164**. In some embodiments, the second coil **164** may be disposed and/or extended at a different direction compared to (e.g., orthogonal to, nearly orthogonal to) the fourth coil **168**. For example, the fourth polygonal conductor **212** and the fifth polygonal conductor **214** of the second coil **164** may be extended along a first direction. In specific embodiments, the seventh polygonal conductor **252** and the eighth polygonal conductor **254** of the fourth coil **168** may be extended along a second direction perpendicular to (e.g., nearly perpendicular to) the first direction.

[0095] In the depicted embodiment, the phase shifter circuitry **68** may receive the input signals at the first terminal **124**. It should be appreciated that in alternative or additional cases, the phase shifter circuitry **68** may receive the input signals at the second terminal **126**. Moreover, the first switch **130** is open, the second switch **132** is open, and the third switch **134** is closed. As such, the fourth polygonal conductor **212** and the fifth polygonal conductor **214** may direct the portion of the input signals in opposite directions based on being cross-coupled and having the figure-eight shape. Furthermore, the fourth switch **136** is open, the fifth switch **138** is open, and the sixth switch **140** is closed. Accordingly, the seventh polygonal conductor **252** and the eighth polygonal conductor **254** may be cross-coupled and have a figure-eight shape to direct the portion of the input signals in opposite directions.

[0096] As mentioned above, directing the input signals in the opposite directions through the adjacent polygonal conductors **212** and **214** of the second coil **164** may reduce induced currents of the first phase shifter circuit **120** during operation. Similarly, in some cases, directing the input signals in the opposite directions through the adjacent polygonal conductors **252** and **254** of the fourth coil **168** may reduce induced currents of the second phase shifter circuit **122** during operation. For example, in specific cases, at least a part of the induced currents of the inductors **146**, **148**, **150**, **152**, **154**, and **156** may destructively combine during operation.

[0097] In some cases, directing the input signals in the opposite directions through the adjacent and polygonal conductors **212** and **214** of the second coil **164** and the adjacent and polygonal conductors **252** and **254** of the fourth coil **168** may reduce induced currents of the phase shifter circuitry **68** during operation. For example, the orthogonal disposition of the polygonal conductors **212** and **214** of the second coil **164** compared to the polygonal conductors **252** and **254** of the fourth coil **168** may increase a distance between the polygonal conductors **212**, **214**, **252** and **254**.

[0098] As such, in some cases, an accumulation of the induced currents of the polygonal conductors **212**, **214**, **252** and **254** may be reduced based on the increased distance between the polygonal conductors **212**, **214**, **252** and **254**. In specific cases, at least a portion of the induced currents of the polygonal conductors **212**, **214**, **252** and **254** may destructively combine to reduce the induced currents of the phase shifter circuitry **68** during operation. Moreover, an undesired effect of the inductors **146**, **148**, **150**, **152**, **154**, and/or **156** on each other or one or more other components of the electronic device **10** including the phase shifter circuitry **68** may be reduced during operation.

[0099] Furthermore, the phase shifter circuitry **68** may have reduced insertion loss based on the reduction of the induced currents. In some cases, the phase shifter circuitry **68** may output signals with improved linearity based on the reduced induced currents. Moreover, the phase shifter circuitry **68** may have a reduced area compared to other phase shifters. For example, the phase shifter circuitry **68** may have a 40 percent, 43 percent, 56 percent, 60 percent, 75 percent, and so on, among other possibilities of percentages of reduced area compared to other phase shifter circuitries.

[0100] It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

[0101] The specific embodiments described above have been shown by way of example, and it should be understood that these embodiments may be susceptible to various modifications and alternative forms. It should be further understood that the claims are not intended to be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.

[0102] The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as "means for [perform]ing [a function] . . . " or "step for [perform]ing [a function] . . . ", it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

What is claimed is:

1. Phase shifter circuitry comprising:

a first phase shifter circuit comprising a first coil and a second coil, the second coil being disposed over and extended around a boundary of the first coil, the second coil comprising a first twisted loop conductor extended in a first direction; and

a second phase shifter circuit comprising a third coil and a fourth coil, the fourth coil being disposed over and enclosed by a boundary of the third coil, the fourth coil comprising a second twisted loop conductor extended in a second direction different than the first direction.

2. The phase shifter circuitry of claim **1**, wherein the first twisted loop conductor and the second twisted loop conductor each comprises a plurality of cross-coupled polygonal conductors forming a figure-eight shape.

3. The phase shifter circuitry of claim **1**, wherein the first phase shifter circuit comprises a first switch configured to couple a first terminal of the first phase shifter circuit to the second phase shifter circuit, and a second switch configured to bypass the second phase shifter circuit.

4. The phase shifter circuitry of claim **3**, wherein processing circuitry coupled to the phase shifter circuitry is configured to close the first switch to bypass the first phase shifter circuit, and close the second switch to bypass the second phase shifter circuit.

5. The phase shifter circuitry of claim **1**, wherein the phase shifter circuitry is configured to adjust a phase of a first signal before being transmitted by an antenna, and adjust a phase of a second signal after being received by the antenna.

6. The phase shifter circuitry of claim **1**, wherein the third coil and the fourth coil do not overlap with the first coil and the second coil.

7. Phase shifter circuitry comprising:

a first coil disposed on a first circuit layer and having a first boundary, comprising

a first polygonal conductor comprising a first inner conductor coupled to a second inner conductor, and

a second polygonal conductor comprising a first outer conductor coupled to a second outer conductor, the first outer conductor being coupled to the first inner conductor, and the second outer conductor coupled to the second inner conductor, the first polygonal conductor being encircled by the second outer conductor,

a second coil disposed on a second circuit layer, the second coil being coupled to the first inner conductor and the second inner conductor, the second coil comprising a first twisted loop conductor having a second boundary that covers the first boundary of the first coil,

a third coil disposed on the first circuit layer, the third coil being coupled to the second outer conductor, the third coil not overlapping with the first coil and the second coil; and

a fourth coil coupled to the third coil, the fourth coil comprising a second twisted loop conductor encircled by the third coil, and the second twisted loop conductor being disposed in a different direction with respect to the first twisted loop conductor.

8. The phase shifter circuitry of claim **7**, wherein the first outer conductor is disposed around the second inner conductor and the second outer conductor is disposed around the first inner conductor.

9. The phase shifter circuitry of claim **7**, wherein the first twisted loop conductor comprises a third polygonal conductor cross-coupled to a fourth polygonal conductor.

10. The phase shifter circuitry of claim **7**, wherein the third coil comprises a third outer conductor and a fourth outer conductor, the fourth coil being coupled to an intersection of the third outer conductor and the fourth outer conductor.

11. The phase shifter circuitry of claim **10**, wherein the fourth coil is disposed on the second circuit layer.

12. The phase shifter circuitry of claim **7**, wherein the second twisted loop conductor comprises a third polygonal conductor cross-coupled to a fourth polygonal conductor.

13. The phase shifter circuitry of claim **7**, wherein the first twisted loop conductor and the second twisted loop conductor each comprises a plurality of interconnected polygonal conductors forming a figure-eight shape.

14. The phase shifter circuitry of claim **7**, wherein the first twisted loop conductor is extended perpendicularly with respect to the second twisted loop conductor.

15. The phase shifter circuitry of claim **7**, comprising a first switch configured to couple a first terminal of the phase shifter circuitry to the third coil bypassing the first coil and the second coil, or a second switch configured to couple a second terminal of the phase shifter circuitry to the first coil bypassing the third coil and the fourth coil.

16. An electronic device comprising:

processing circuitry;

an antenna; and

phase shifter circuitry coupled to the antenna and the processing circuitry, the phase shifter circuitry comprising

a first coil,

a second coil coupled to the first coil, the second coil being disposed over and extending around the first coil, the second coil comprising a first twisted loop conductor extended in a first direction,

a third coil coupled to the first coil, the third coil not overlapping with the first coil and the second coil, and

a fourth coil coupled to the third coil, the fourth coil being disposed over and surrounded by the third coil, the fourth coil comprising a second twisted loop conductor extending in a second direction different than the first direction.

17. The electronic device of claim **16**, comprising a first switch configured to couple a first terminal of the phase shifter circuitry to the third coil bypassing the first coil and the second coil, and a second switch configured to couple a second terminal of the phase shifter circuitry to the first coil bypassing the third coil and the fourth coil.

18. The electronic device of claim **17**, wherein the processing circuitry is configured to close the first switch to bypass the first coil and the second coil, and close the second switch to bypass the third coil and the fourth coil.

19. The electronic device of claim **16**, wherein the phase shifter circuitry is configured to adjust a phase of a first signal before being transmitted by the antenna, and adjust a phase of a second signal before being received by the antenna.

20. The electronic device of claim **19**, wherein the processing circuitry is configured to generate the first signal, and receive the second signal.

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